SUTOLIMIN WIDJAJA

(+65) 8365-1042 | sutolimin.45@gmail.com | LinkedIn | GitHub | Portfolio | Student Pass holder

Education

Singapore Management University Aug 2025 – May 2028 (Expected)

B.S. in Software Engineering (Work-Study Degree)

- **Relevant Coursework**: Operating Systems and Networking, Web Application Development, Algorithms and Programming, Data Management, Linear Algebra, Introduction to Probability and Statistics, Data Structures & Algorithms
- **Profiles**: Codeforces Specialist, 1500+ (Profile), LeetCode 1850+ (Profile)
- **CCAs**: Competitive Programming, Artificial Intelligence, Cybersecurity, Quantitative Finance
- **Working Rights**: Student Visa

Work Experience

Singapore Management University Aug 2026 – Present

Teaching Assistant | Algorithms and Programming *Singapore, Singapore*

- Teach weekly lab sessions for 35 students on recursion, sorting, complexity analysis, and graph theory, walking through problem-solving approaches.

Algoverse Jul 2026 – Oct 2026

Undergraduate Researcher | Machine Learning, NLP, LLM Evaluation *Singapore, Singapore*

- Built an LLM-as-judge pipeline to evaluate 5 compression policies for multi-agent LLM memory on a 30,000-message benchmark.
- Showed that keeping the right evidence at a 25% memory budget restored accuracy by up to 75pp ($p \leq 0.001$).

PreciX Jul 2026 – Aug 2026

Machine Learning Intern | Python, PyTorch, scikit-learn *Singapore, Singapore*

- Engineered preprocessing and validation pipelines for knee-kinematics time-series data from multi-sensor wearables.
- Integrated Prefect to monitor pipeline runs, exporting metrics via Prometheus to Grafana dashboards for observability.
- Trained and evaluated machine learning models to identify biomechanical markers of ACL injury and track patient recovery.

FlyRank AI Jul 2026 – Aug 2026

Machine Learning Intern | Python, PyTorch *Singapore, Singapore*

- Built a PyTorch model forecasting 2026 FIFA World Cup match outcomes from engineered team- and match-level features.
- Developed reproducible preprocessing, training, and evaluation pipelines for a 12-week ML capstone.

Columns May 2026 – Jul 2026

Software Engineering Intern *Singapore, Singapore*

- Shipped features to the production web, iOS, and Android clients of columns.app, supporting 10,000+ concurrent users across live App Store and Google Play releases.
- Resolved production crashes and user-facing defects across React web and mobile clients by triaging Sentry error reports, reproducing root causes, and verifying fixes with regression testing.
- Reduced release regressions by validating every build through GitHub Actions CI while shipping UI and workflow improvements.

Projects

Limit Order Book | *C++, Ninja, LOBSTER* | GitHub Apr 2026 – Present

- Built a C++ limit order book with AVL-tree price levels and FIFO order queues for price-time priority, sustaining ∼7M operations/sec across 4 sharded books.

Movie Recommendation System | *C++, Ninja, MovieLens* | GitHub May 2026 – Present

- Built a scalable collaborative filtering recommendation system using a Compressed Sparse Row (CSR) sparse matrix structure for memory-efficient storage of the user-item rating matrix.
- Benchmarked system performance against the MovieLens 100K dataset, achieving an RMSE of 0.91.
- Parallelized similarity-matrix construction with a custom thread pool and built a title search that retrieves the 10 most similar films for a query.

Real-Time Chat Application | *React.js, Next.js, TypeScript, PostgreSQL, Socket.IO* | GitHub | Link May 2026 – Present

- Architected a full-stack, real-time messaging platform, decoupling the Next.js frontend on Vercel from a persistent Socket.IO event server to enable sub-100ms bidirectional communication over WebSockets.
- Applied modular software design to the Express backend, separating HTTP routes from data-access code, and built the PostgreSQL schema through 5 ordered SQL migrations run by a custom script.

Achievements & Activities

- **Silver Medallist**, International Junior Cybersecurity Olympiad
- **3rd Place**, GMI x Z.ai Agent Forge AI Hackathon
- **3rd Place**, TicTacCode (Algorithms & CTF), SMU Ellipsis
- **Top 5**, Tencent AI for Good Hackathon
- **Top 8**, AI Tinkerers SG x Tencent Cloud Hackathon
- **Top 12**, GIC Code to Impact 2026

Technical Skills

Languages: C/C++, Java, Python, JavaScript/TypeScript, HTML5/CSS3, Bash (Linux)
Frameworks: Vue.js, Next.js, React.js, Node.js, Express.js, REST APIs
Libraries: Matplotlib, NumPy, Pandas, PyTorch, scikit-learn
Databases: PostgreSQL, MySQL, MongoDB, Redis
Tools/Technologies: Git, Vim, Postman, GitHub, Google Cloud, AWS, Docker, Linux/Unix, Sentry, Jira, Maestro